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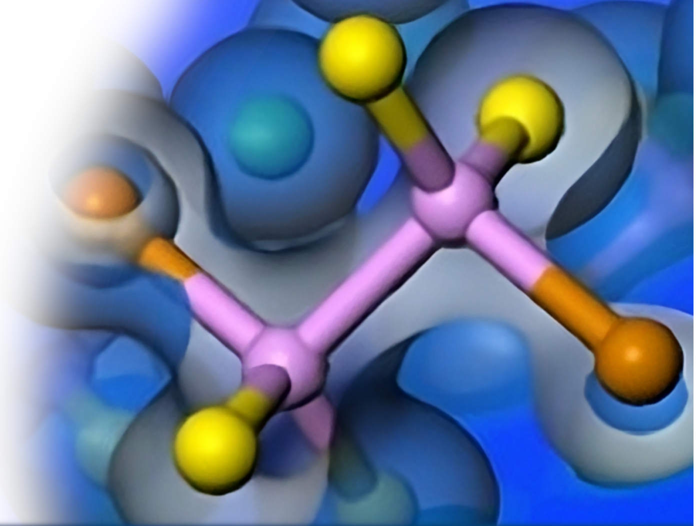

Reciprocal-space, DOS & dispersion

Phil Hasnip, University of York, UK

Recap of Bloch's Theorem

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Bloch's theorem: in a periodic potential, the density has the same periodicity. The possible wavefunctions are all 'quasi-periodic':

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}} u_k(\mathbf{r}).$$

We write $u_k(\mathbf{r})$ in a plane-wave basis as:

$$u_k(\mathbf{r}) = \sum_G c_{Gk} e^{i\mathbf{G}.\mathbf{r}},$$

where $\mathbf{G}$ are the reciprocal lattice vectors, defined so that $\mathbf{G}.\mathbf{L} = 2\pi m$.

First Brillouin Zone

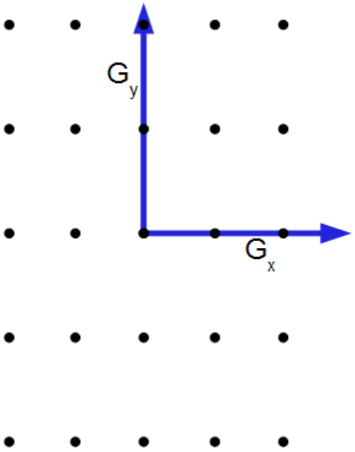
Adding or subtracting **G** from **k** leaves properties unchanged – system is periodic in reciprocal-space too.

Only need to study behaviour in the reciprocal-space unit cell, to know how it behaves everywhere.

Conventional to consider unit cell surrounding $\mathbf{G} = 0$, called the first Brillouin zone.

First Brillouin Zone (2D)

The region of reciprocal space nearer to the origin than any other allowed wavevector is called the 1st Brillouin zone.



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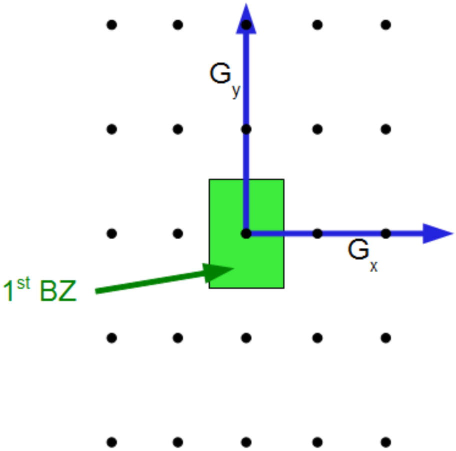

E versus $\mathbf{k}$

How does the energy of states vary across the Brillouin zone? Let's consider one particular wavefunction:

$$\psi(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}}u(\mathbf{r})$$

We'll look at two different limits – electrons with high potential energy, and electrons with high kinetic energy.

Very localised electrons

If an electron is trapped in a very strong potential, then we can neglect the kinetic energy and write:

$$\hat{H} = \hat{V}$$

The energy of our wavefunction is then

$$\begin{aligned} E(k) &= \int \psi^\star(\mathbf{r})V(\mathbf{r})\psi(\mathbf{r})d^3\mathbf{r} \\ &= \int V(\mathbf{r})|\psi(\mathbf{r})|^2 d^3\mathbf{r} \\ &= \int V(\mathbf{r})|u(\mathbf{r})|^2 d^3\mathbf{r} \end{aligned}$$

It doesn't depend on **k** at all! We may as well do everything at $\mathbf{k} = 0$.

H band structure

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Free Electrons

For an electron moving freely in space there is no potential, so the Hamiltonian is just the kinetic energy operator:

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2$$

The eigenstates of the Hamiltonian are just plane-waves – i.e. $c_{Gk} = 0$ except for one particular **G**.

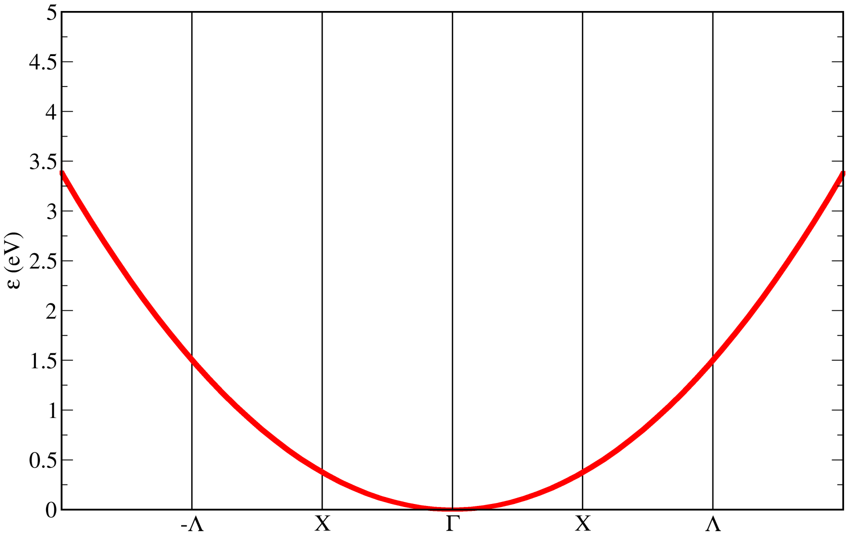
Our wavefunction is now

$$\begin{aligned} \psi(\mathbf{r}) &= c_G e^{i(\mathbf{k}+\mathbf{G}).\mathbf{r}} \\ \Rightarrow \nabla^2\psi(\mathbf{r}) &= -(\mathbf{k}+\mathbf{G})^2\psi(\mathbf{r}) \end{aligned}$$

$$
\begin{aligned}
E(\mathbf{k}) &= -\frac{\hbar^2}{2m}\int \psi^\star(\mathbf{r})\nabla^2\psi(\mathbf{r})d^3\mathbf{r} \\
&= \frac{\hbar^2}{2m}(\mathbf{k}+\mathbf{G})^2\int \psi^\star(\mathbf{r})\psi(\mathbf{r})d^3\mathbf{r} \\
&= \frac{\hbar^2}{2m}(\mathbf{k}+\mathbf{G})^2
\end{aligned}
$$

So $E(\mathbf{k})$ is quadratic in $\mathbf{k}$, with the lowest energy state $\mathbf{G}=0$.

Free Electrons



Each state has an energy that changes with **k** – they form energy bands in reciprocal space.

Recall that the energies are periodic in reciprocal-space – there are parabolae centred on each of the reciprocal lattice points.

Free Electrons

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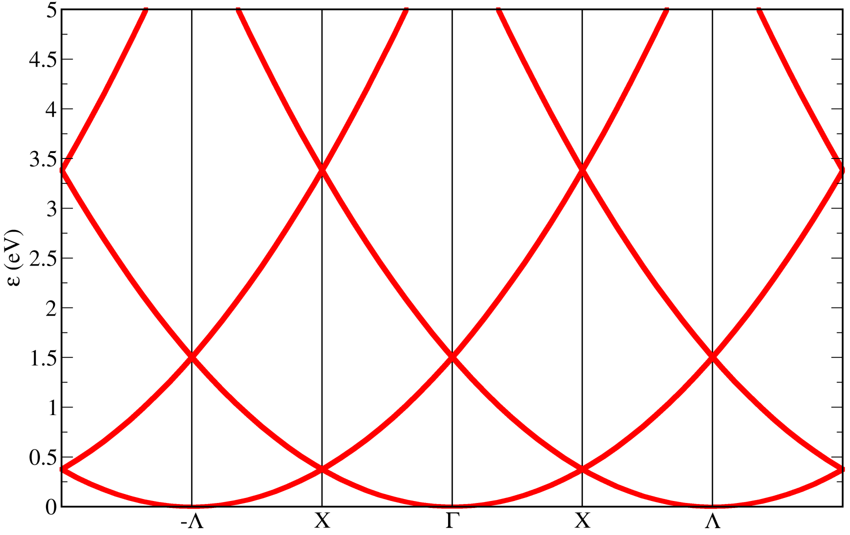
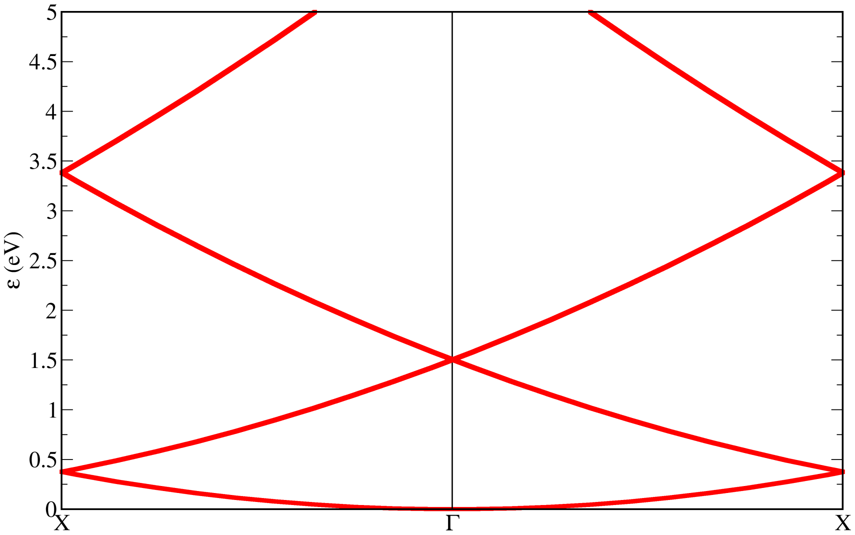
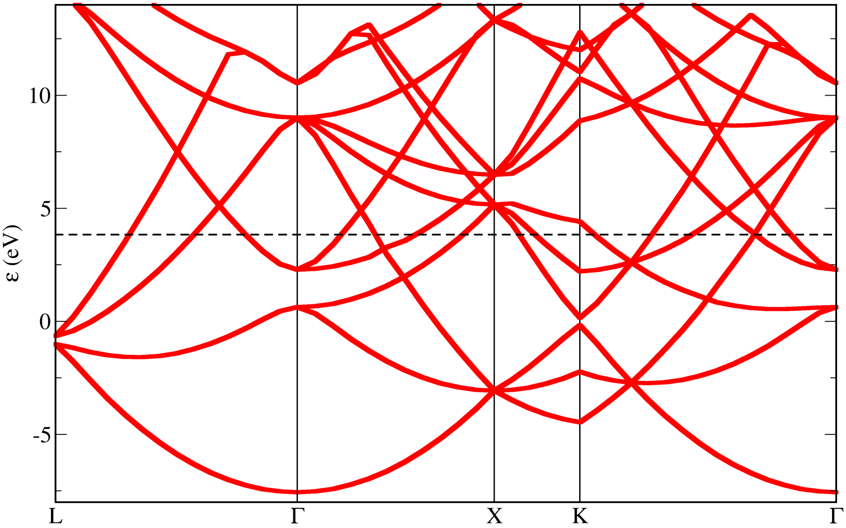


All of the information we need is actually in the first Brillouin zone, so it is conventional to concentrate on that.

Free Electrons



Al band structure



In 3D things get complicated. In general the reciprocal lattice vectors do not form a simple cubic lattice, and the Brillouin zone can have all kinds of shapes.

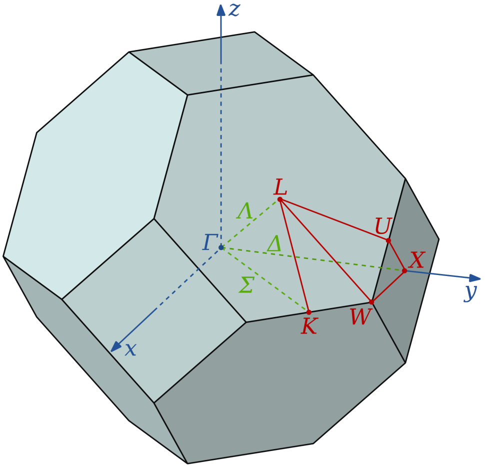


Image from en.wikipedia.org/wiki/Empty_lattice_approximation

Band structure

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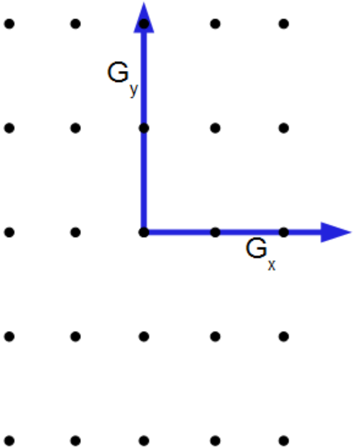
The way the energies of all of the states change with **k** is called the band structure.

k is a 3D vector; easier to plot energies along special high-symmetry directions. These energies represent either maximum or minimum energies for the bands across the whole Brillouin zone.

In real materials electrons are neither completely localised nor completely free, but we can still see those characteristics in band structures.


G_y
G_x

Band Structure k-points (2D)

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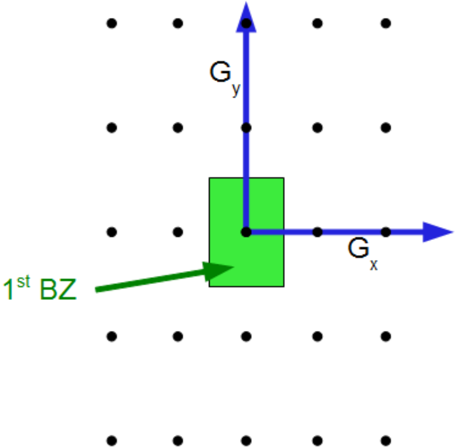
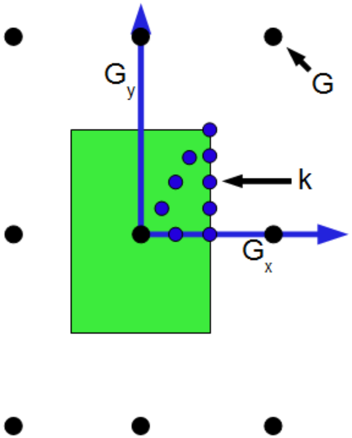


Band Structure k-points (2D)



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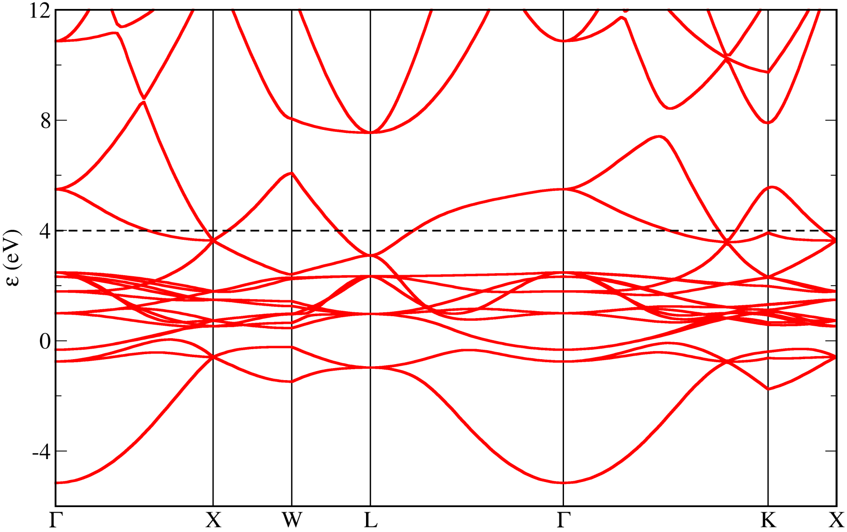

Transitions

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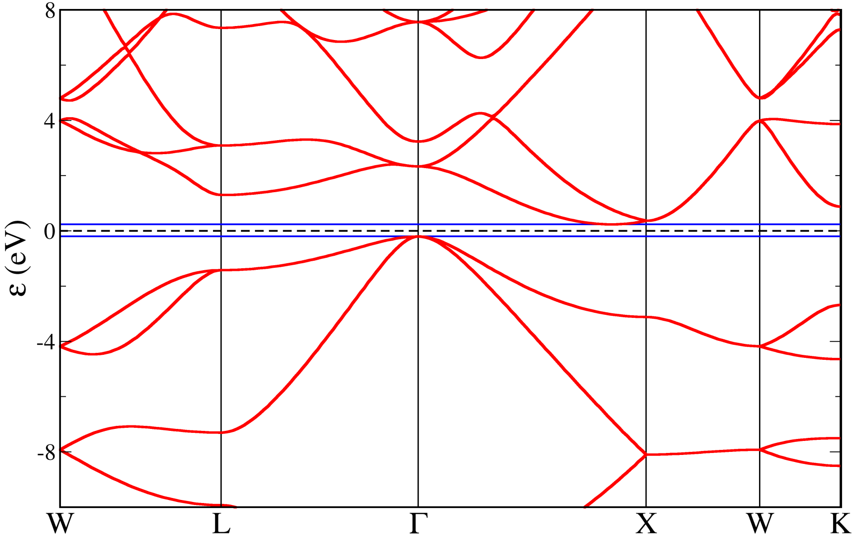
- Lowest N_e states are occupied by electrons
- At $T = 0$K there is an energy below which all states are occupied, and above which all states are empty; this is the Fermi energy, E_F
- Band-structures often shifted so $E_F = 0$

In semi-conductors and insulators there is a region of energy just above E_F with no bands – the band gap.

Band structure of Si



The band structure is a good way to visualise the wavevector-dependence of the energy states, the band-gap, and the possible electronic transitions.

The actual transition probability depends on how many states are available in both the initial and final energies. The band structure is not a reliable guide here, since it only tells you about the bands along high symmetry directions.

Densities of States

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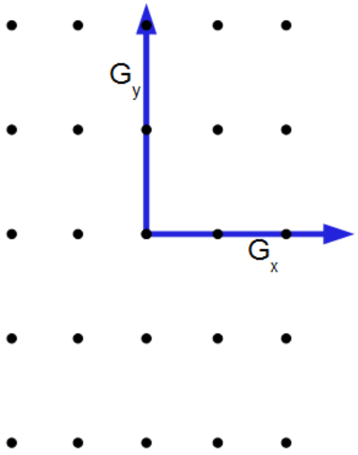
What we need is the full density of states across the whole Brillouin zone, not just the special directions.

The density of states $g(E)$ is,

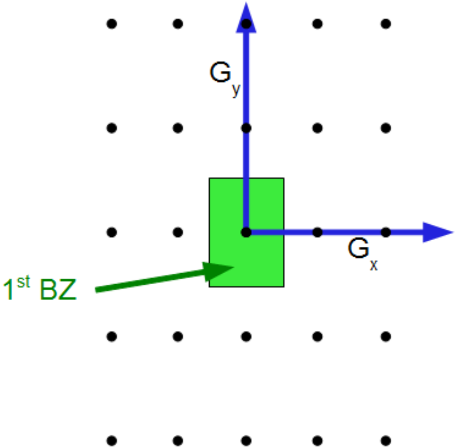
$$g(E) = \frac{1}{\Omega_G} \frac{dN}{dE}$$

where Ω_G is the volume of reciprocal space associated with each **G**-vector.

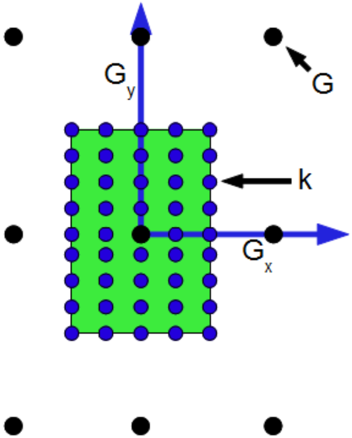
We have to sample the Brillouin zone evenly, just as we do for the calculation of the ground state.


G_y
G_x

Densities of States



Densities of States

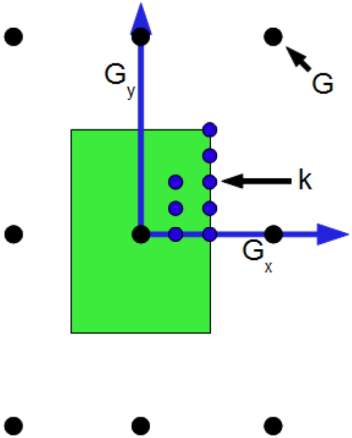


Densities of States

Often the crystal will have extra symmetries which reduce the number of **k**-points we have to sample.

Once we've applied all of the relevant symmetries to reduce the **k**-points required, we are left with the *irreducible wedge*.

Densities of States



For free electrons

$$E(\mathbf{k}) \propto (\mathbf{k}+\mathbf{G})^2$$

So for a fixed E the $\mathbf{k}$-vectors form a spherical shell of area $4\pi|\mathbf{k}+\mathbf{G}|^2$.

$$\begin{aligned} \frac{dN}{dk} &\propto |\mathbf{k}+\mathbf{G}|^2 \\ \Rightarrow g(E) &= \frac{1}{\Omega_G}\frac{dN}{dE} = \frac{1}{\Omega_G}\frac{dN}{dk}\frac{dk}{dE} \\ &\propto |\mathbf{k}+\mathbf{G}|^2\frac{1}{|\mathbf{k}+\mathbf{G}|} \\ &\propto \sqrt{E} \end{aligned}$$

Free electrons

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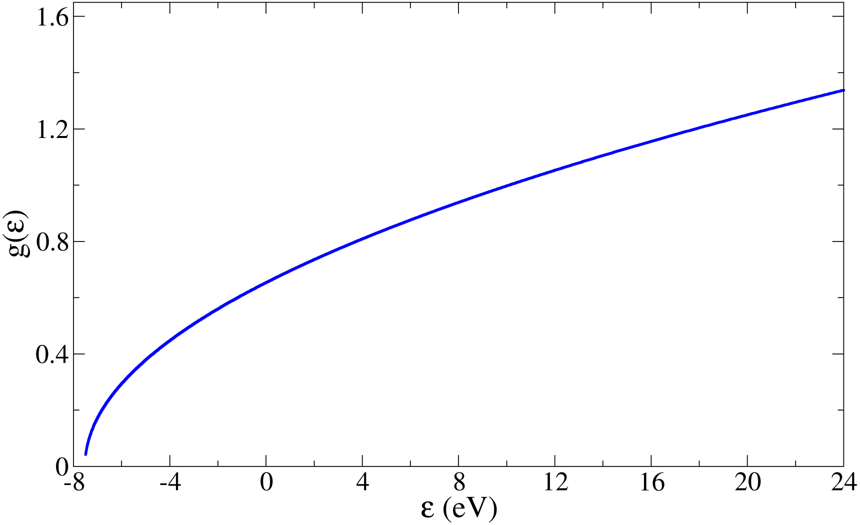

Aluminium

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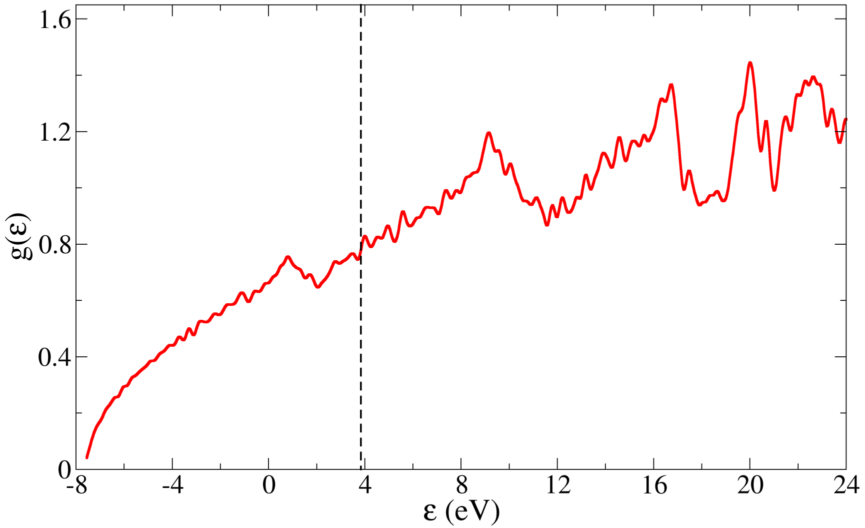

Aluminium

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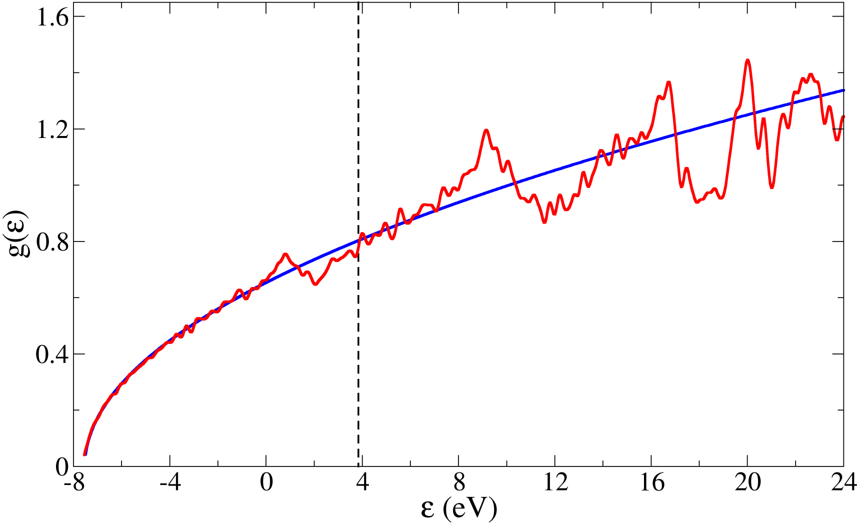
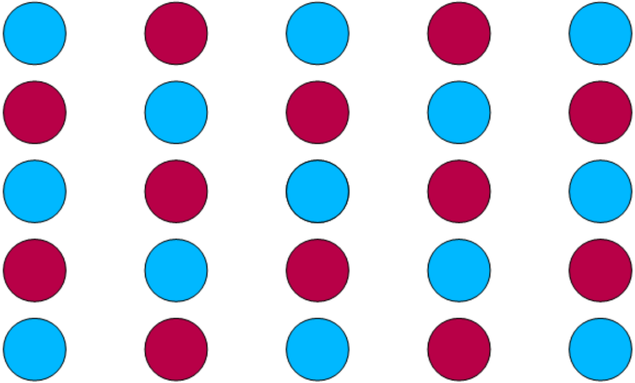


Computing a band structure or a DOS is straightforward:

- Compute the ground state density with a good **k**-point sampling
- Fix the density, and find the states at the band structure/DOS **k**-points

Because the density is fixed for the band structure/DOS calculation itself, it can be quite a lot quicker than the ground state calculation even though it may have more **k**-points.

When a vibration travels through a crystal, it creates a periodic distortion to the atoms.



Phonons

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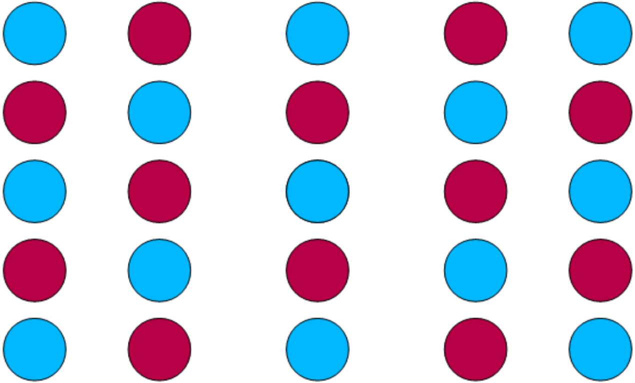
When a vibration travels through a crystal, it creates a periodic distortion to the atoms.



Phonons

Periodic distortion $\longrightarrow$ wavevector **q**.
Periodicity not required to be the same as the unit cell.

Atoms are displaced from equilibrium positions by the vibration. The displacement d is real, not complex, so:

$$d_q(\mathbf{r}) = a_q \cos(\mathbf{q}.\mathbf{r})$$

The natural vibrations of a material are called *phonons*.

Phonon energies depend on **q**: plot *phonon* band structure (usually as frequency $\omega(\mathbf{q})$ against **q**, where $E = \hbar\omega$).

Shows frequency of different lattice vibrations, from long-wavelength acoustic modes to the shorter optical ones.

Si phonon bandstructure

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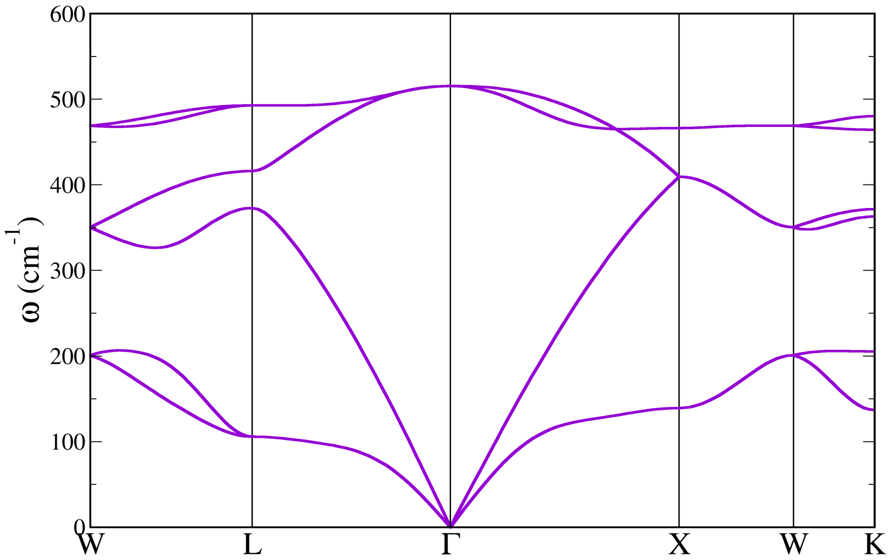
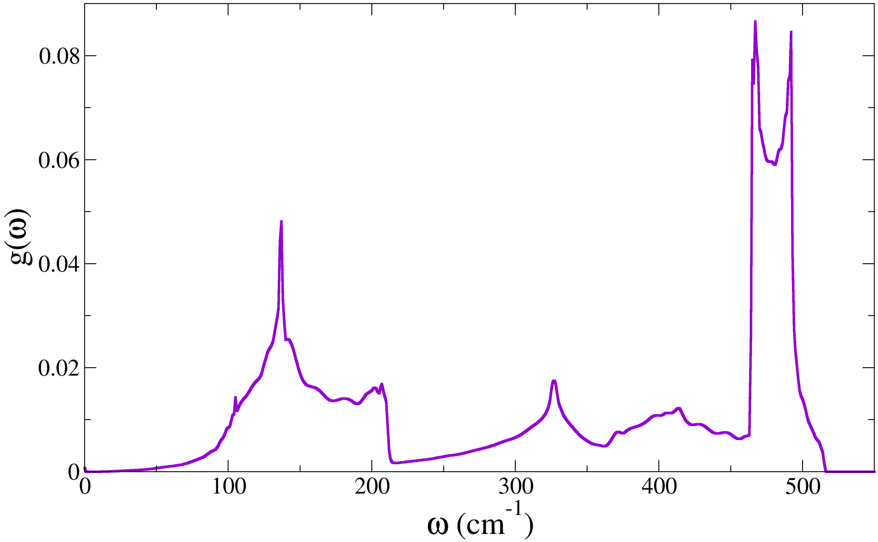


Si phonon DOS



Summary

Band structures are useful ways to illustrate how the band (or phonon) energies change across the 3D Brillouin zone, with a 1D plot. They show E against **k** (or ω against **q**) along the high-symmetry directions.

Density-of-states plots are obtained by sampling the *entire* Brillouin zone, and show you how many states there are for each E (or ω).